

RFID-Based Student Fee and Attendance Management System

The RFID-Based Student Fee and Attendance Management System is an IoT-enabled solution designed to automate and streamline attendance tracking and fee management in educational institutions. The system utilizes RFID cards assigned to students, which, when scanned at designated readers, automatically record attendance and retrieve fee payment details. A backend system powered by NodeMCU ESP8266 and MongoDB ensures real-time data processing and updates.

By integrating RFID technology, the system minimizes manual errors, reduces administrative workload, and prevents overcrowding at fee counters by notifying students of their pending dues in advance. The cashier updates fee payment records in real time, eliminating the need for repeated scanning. Additionally, attendance tracking is automated as students enter classrooms, ensuring accurate record-keeping.

This cost-effective and scalable solution enhances institutional efficiency by providing instant access to student records, improving transparency, and ensuring a seamless experience for both students and administrators. The project highlights the effective use of IoT and database technologies to optimize operational processes in educational environments.